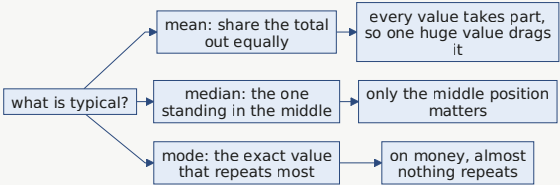


typical, spread and denominators

Week 1, Day 4. Same eight panels with about a quarter of the cells removed.

1 the three "typical" numbers



	Answers	Reach for it when
Mean	_____	The shape is roughly even
Median	The record in the middle of the queue	_____
Mode	The value that repeats most	_____

Crux. nobody is right, they answer _____, and the asker rarely says which one they meant.

2 the one-line skew test

_____ / _____		
Ratio	Shape	Send
Close to 1	_____	Either
Well above 1	Right tail, something large	_____
Well below 1	_____	Median
Run it first on any column you have never seen.		

3 what one record does

seven orders	mean Rs 1,676.43	median Rs 1,310
swap the top for 480000	mean Rs _____	median Rs _____
Crux. the mean _____, the median moved by _____. Not "a little".		

4 the fence

```
q1, _, q3 = statistics.quantiles(values, n=4)
iqr = _____
upper = _____
lower = _____
```

Crux. a fence is a _____, never a _____. An outlier is a finding to investigate before it is a row to delete.

5 grouping without knowing the categories

```
buckets = {}
for r in orders:
    key = r[field]
    if _____:
        buckets[key] = {"count": 0, "amounts": [], "returned": 0}
    buckets[key]["count"] += 1
    buckets[key]["amounts"]._____ (r["amount"])
    if r["status"] == "_____":
        buckets[key]["returned"] += 1
```

Short form of the same idea: counts[key] = counts._____ (key, _____) + 1

Crux. every hard-coded list of categories is _____. Hard-code three and Kalpa has four, and you get _____: '_____'.

6 count first, then median

Accumulate one record at a time	Needs the whole group first
count, sum, returned count, _____,	_____, _____, the fence

Crux. there is no _____.	

7 the honest sentence

[_____] [_____] [_____]
Write the sentence for Business, which had 1 returned of 9 orders:

Crux. the third part is the one that goes missing between your notebook and somebody else's slide.

8 before you send a rate

Check	If it fails
_____	It is not ready
Any group under your stated floor	_____
Ranked groups of very different sizes	Say what the ranking is partly measuring
A mean quoted on a money column	_____
A difference being called real	_____

THE THREE LINES WORTH MEMORISING

- 1. The mean was _____ and the description was _____.
- 2. On a money column, send the _____, and say that is what you sent.
- 3. Every rate carries its _____, or it lies for you while you are not in the room.

THE WORDS ON THIS SHEET

- Mean** The total shared out equally, which every value takes part in
- Median** The value standing in the middle of the sorted column
- Mode** The exact value that repeats most, which on money repeats twice in 44
- Range** Maximum minus minimum, built from the two least typical values

- Interquartile range** The spread of the middle half, and the basis of the fence
- Skew** Which side of the median the tail is on
- Denominator** The count a rate rests on, which travels with it
- Trust floor** A record count below which you decline to rank, chosen and defended